

Euler's Identity: The Most Beautiful Equation in Mathematics

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Abstract

This project investigates the most popular mathematical equation, the Euler Identity, $e^{i\pi} + 1 = 0$. We consider the five basic constants, (e , i , π , 0, and 1), and we determine the identity of these numbers using Taylor series and complex analysis, we explain the historical meaning of each of these constants and we also prove the extensive interrelations between the apparently unrelated areas of mathematics.

1 Introduction

Euler's Identity, $e^{i\pi} + 1 = 0$, is hailed as one of the most beautiful mathematics equations. It remarkably relates between five constants of mathematics:

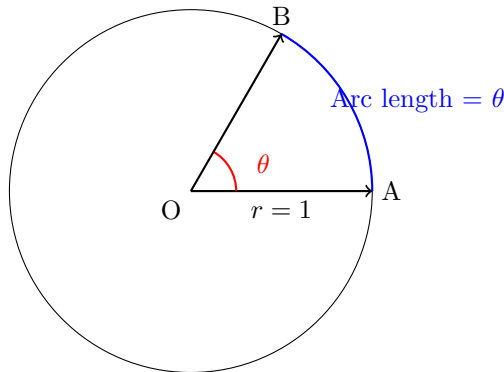
- 0: The additive identity
- 1: The multiplicative identity
- π : The circle constant
- e : The natural exponential base
- i : The imaginary unit

Such a formula seems on the face of it like a riddle, how could an irrational number to the imaginary power of an irrational number should give a simple negative integer? This mystery is solved in this project in both mathematical derivation and context.

2 The Five Fundamental Constants

2.1 π : The Circle Constant

π is the ratio of a circumference of a circle to its diameter which is about 3.14159. In higher mathematics, angles are normally expressed in radians as opposed to degrees where a complete circle measures 2π radians. This is of great significance since in a unit circle, the arc length is the same as the angle in radians.



$$\text{Circumference} = 2\pi r = 2\pi \text{ (when } r = 1\text{)}$$

The diagram above shows a unit circle (radius = 1). In case an angle θ is expressed in radians, the arc length is equal to θ . Such an inherent connection is that radians are the unit of choice in calculus and higher mathematics.

2.2 i : The Imaginary Unit

The imaginary unit i is defined as $i = \sqrt{-1}$, solving $x^2 + 1 = 0$. Complex numbers extend real numbers, enabling solutions to previously unsolvable equations. The powers of i follow a cyclic pattern:

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad \dots$$

Historical Emergence from Cubic Equations

The need for i emerged practically when 16th-century mathematicians solved cubic equations. To find *real solutions*, they often needed intermediate steps involving $\sqrt{-1}$.

In 1572, Rafael Bombelli solved $x^3 - 15x - 4 = 0$. The formula gave:

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}},$$

yet the real root was $x = 4$. Bombelli treated $\sqrt{-1}$ algebraically, noticing:

$$(2 + \sqrt{-1})^3 = 2 + \sqrt{-121} \quad \text{and} \quad (2 - \sqrt{-1})^3 = 2 - \sqrt{-121}.$$

Thus, $x = (2 + \sqrt{-1}) + (2 - \sqrt{-1}) = 4$, showing imaginary numbers as a bridge to real solutions.

Euler's Notation

Leonhard Euler popularized the symbol i for $\sqrt{-1}$ in 1770. His formula $e^{i\theta} = \cos \theta + i \sin \theta$ led directly to Euler's identity $e^{i\pi} + 1 = 0$.

2.3 e : The Natural Exponential Base

The constant e (approximately 2.71828) is the unique number for which the function $f(x) = e^x$ equals its own derivative. This property makes e fundamental in calculus, differential equations, and exponential growth modeling.

The discovery and derivation of e involves examining the derivative of exponential functions and finding the special base where the function equals its own derivative. This derivation will be presented in Section 4.2 using Taylor series, which provides an elegant method to understand this remarkable property.

2.4 0 and 1: Additive and Multiplicative Identities

- 0: The additive identity, where $a + 0 = a$ for any number a
- 1: The multiplicative identity, where $a \times 1 = a$ for any number a

These two constants form the foundation of arithmetic and algebra.

3 Taylor Series Expansion

3.1 Introduction to Taylor Series

Taylor series represent functions as infinite polynomials, allowing us to approximate complex functions with polynomials of increasing accuracy. The general Taylor series expansion of a function $f(x)$ about a point a is given by:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

When we center the expansion at $a = 0$, we obtain a special case called the Maclaurin series:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

The Maclaurin series is essentially a Taylor series centered at zero. For many elementary functions, the Maclaurin series provides the most convenient representation.

3.2 Taylor Series for Key Functions

3.2.1 Exponential Function e^x

Let us derive the Taylor series for e^x by assuming it can be expressed as an infinite polynomial:

$$e^x = a + bx + cx^2 + dx^3 + ex^4 + fx^5 + gx^6 + hx^7 + \cdots$$

The key insight is that the exponential function has the remarkable property that it is its own derivative:

$$\frac{d}{dx}[e^x] = e^x$$

1. **Step 1:** Set $x = 0$:

$$e^0 = a + b(0) + c(0)^2 + d(0)^3 + \cdots = a$$

Since $e^0 = 1$, we get $a = 1$.

2. **Step 2:** Differentiate both sides:

$$\frac{d}{dx}[e^x] = e^x = b + 2cx + 3dx^2 + 4ex^3 + 5fx^4 + 6gx^5 + 7hx^6 + \cdots$$

Since the derivative of e^x is e^x itself, and we already have the series representation, we can set $x = 0$:

$$e^0 = b = 1 \Rightarrow b = 1$$

3. **Step 3:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 2c + 6dx + 12ex^2 + 20fx^3 + 30gx^4 + 42hx^5 + \cdots$$

Set $x = 0$:

$$e^0 = 2c = 1 \Rightarrow c = \frac{1}{2}$$

4. **Step 4:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 6d + 24ex + 60fx^2 + 120gx^3 + 210hx^4 + \cdots$$

Set $x = 0$:

$$e^0 = 6d = 1 \Rightarrow d = \frac{1}{6}$$

5. **Step 5:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 24e + 120fx + 360gx^2 + 840hx^3 + \cdots$$

Set $x = 0$:

$$e^0 = 24e = 1 \Rightarrow e = \frac{1}{24}$$

6. **Step 6:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 120f + 720gx + 2520hx^2 + \dots$$

Set $x = 0$:

$$e^0 = 120f = 1 \Rightarrow f = \frac{1}{120}$$

7. **Step 7:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 720g + 5040hx + \dots$$

Set $x = 0$:

$$e^0 = 720g = 1 \Rightarrow g = \frac{1}{720}$$

8. **Step 8:** Differentiate again:

$$\frac{d}{dx}[e^x] = e^x = 5040h + \dots$$

Set $x = 0$:

$$e^0 = 5040h = 1 \Rightarrow h = \frac{1}{5040}$$

Now let's examine the coefficients we've found:

$$\begin{aligned} a &= 1 = \frac{1}{0!} \\ b &= 1 = \frac{1}{1!} \\ c &= \frac{1}{2} = \frac{1}{2!} \\ d &= \frac{1}{6} = \frac{1}{3!} \\ e &= \frac{1}{24} = \frac{1}{4!} \\ f &= \frac{1}{120} = \frac{1}{5!} \\ g &= \frac{1}{720} = \frac{1}{6!} \\ h &= \frac{1}{5040} = \frac{1}{7!} \end{aligned}$$

We observe a clear pattern: the coefficient of x^n is $\frac{1}{n!}$. Thus, we obtain the Taylor series for e^x :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

This derivation beautifully demonstrates why e^x is its own derivative. When we differentiate the series term by term:

$$\frac{d}{dx}e^x = \frac{d}{dx} \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right) = 0 + 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = e^x$$

each term transforms into the previous term in the series, perfectly preserving the original function.

3.2.2 Discovery of e through Taylor Series

The Taylor series provides an elegant way to understand the special property of e . If we set $x = 1$ in the Taylor series for e^x , we get:

$$e = e^1 = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots$$

Calculating the first few terms:

$$1 \text{ term: } 1 = 1$$

$$2 \text{ terms: } 1 + 1 = 2$$

$$3 \text{ terms: } 2 + \frac{1}{2} = 2.5$$

$$4 \text{ terms: } 2.5 + \frac{1}{6} \approx 2.6667$$

$$5 \text{ terms: } 2.6667 + \frac{1}{24} \approx 2.7083$$

$$6 \text{ terms: } 2.7083 + \frac{1}{120} \approx 2.7167$$

$$7 \text{ terms: } 2.7167 + \frac{1}{720} \approx 2.7181$$

This series converges to $e \approx 2.71828$. This is how this constant e was derived.

3.2.3 Sine Function

Let us derive the Taylor series for $\sin x$ by assuming it can be expressed as an infinite polynomial:

$$\sin x = a + bx + cx^2 + dx^3 + ex^4 + fx^5 + gx^6 + hx^7 + \dots$$

1. **Step 1:** Set $x = 0$:

$$\sin 0 = a + b(0) + c(0)^2 + d(0)^3 + \dots = a$$

Since $\sin 0 = 0$, we get $a = 0$.

2. **Step 2:** Differentiate both sides:

$$\frac{d}{dx}[\sin x] = \cos x = b + 2cx + 3dx^2 + 4ex^3 + 5fx^4 + 6gx^5 + 7hx^6 + \dots$$

Set $x = 0$:

$$\cos 0 = b = 1 \Rightarrow b = 1$$

3. **Step 3:** Differentiate again:

$$\frac{d}{dx}[\cos x] = -\sin x = 2c + 6dx + 12ex^2 + 20fx^3 + 30gx^4 + 42hx^5 + \dots$$

Set $x = 0$:

$$-\sin 0 = 2c = 0 \Rightarrow c = 0$$

4. **Step 4:** Differentiate again:

$$\frac{d}{dx}[-\sin x] = -\cos x = 6d + 24ex + 60fx^2 + 120gx^3 + 210hx^4 + \dots$$

Set $x = 0$:

$$-\cos 0 = 6d = -1 \Rightarrow d = -\frac{1}{6}$$

5. **Step 5:** Differentiate again:

$$\frac{d}{dx}[-\cos x] = \sin x = 24e + 120fx + 360gx^2 + 840hx^3 + \dots$$

Set $x = 0$:

$$\sin 0 = 24e = 0 \Rightarrow e = 0$$

6. **Step 6:** Differentiate again:

$$\frac{d}{dx}[\sin x] = \cos x = 120f + 720gx + 2520hx^2 + \dots$$

Set $x = 0$:

$$\cos 0 = 120f = 1 \Rightarrow f = \frac{1}{120}$$

Continuing this process, we observe a pattern: only odd-powered terms remain, and the coefficients alternate in sign with denominators that are factorials of odd numbers. Thus, we obtain:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

3.2.4 Cosine Function

Similarly, let us derive the Taylor series for $\cos x$ by assuming:

$$\cos x = a + bx + cx^2 + dx^3 + ex^4 + fx^5 + gx^6 + hx^7 + \dots$$

1. **Step 1:** Set $x = 0$:

$$\cos 0 = a = 1 \Rightarrow a = 1$$

2. **Step 2:** Differentiate both sides:

$$\frac{d}{dx}[\cos x] = -\sin x = b + 2cx + 3dx^2 + 4ex^3 + 5fx^4 + 6gx^5 + 7hx^6 + \dots$$

Set $x = 0$:

$$-\sin 0 = b = 0 \Rightarrow b = 0$$

3. **Step 3:** Differentiate again:

$$\frac{d}{dx}[-\sin x] = -\cos x = 2c + 6dx + 12ex^2 + 20fx^3 + 30gx^4 + 42hx^5 + \dots$$

Set $x = 0$:

$$-\cos 0 = 2c = -1 \Rightarrow c = -\frac{1}{2}$$

4. **Step 4:** Differentiate again:

$$\frac{d}{dx}[-\cos x] = \sin x = 6d + 24ex + 60fx^2 + 120gx^3 + 210hx^4 + \dots$$

Set $x = 0$:

$$\sin 0 = 6d = 0 \Rightarrow d = 0$$

5. **Step 5:** Differentiate again:

$$\frac{d}{dx}[\sin x] = \cos x = 24e + 120fx + 360gx^2 + 840hx^3 + \dots$$

Set $x = 0$:

$$\cos 0 = 24e = 1 \Rightarrow e = \frac{1}{24}$$

Continuing this process, we observe that only even-powered terms remain, and the coefficients alternate in sign with denominators that are factorials of even numbers. Thus, we obtain:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

4 Derivation of Euler's Formula

4.1 Pattern Recognition in Taylor Series

Let's examine the Taylor series we derived:

$$\begin{aligned}e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \cdots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\end{aligned}$$

Notice the similarity between these series. The exponential series contains all the terms from both the cosine and sine series, but with all positive signs. The cosine series has the even powered terms from the exponential series but with alternating signs. The sine series has the odd powered terms from the exponential series, also with alternating signs.

4.2 Introducing the Imaginary Unit

Now consider what happens when we replace x with ix in the exponential series:

$$\begin{aligned}e^{ix} &= 1 + (ix) + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \frac{(ix)^7}{7!} + \cdots \\ &= 1 + ix + \frac{i^2 x^2}{2!} + \frac{i^3 x^3}{3!} + \frac{i^4 x^4}{4!} + \frac{i^5 x^5}{5!} + \frac{i^6 x^6}{6!} + \frac{i^7 x^7}{7!} + \cdots\end{aligned}$$

Using the cyclic pattern of powers of i :

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad i^6 = -1, \quad i^7 = -i, \dots$$

Substituting these values:

$$e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \frac{x^6}{6!} - \frac{ix^7}{7!} + \cdots$$

Now let's separate the real and imaginary parts:

$$e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right) + i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)$$

Recognizing the Taylor series we derived earlier:

$$e^{ix} = \cos x + i \sin x$$

This is Euler's Formula, one of the most important relationships in mathematics.

4.3 Obtaining Euler's Identity

Substituting $x = \pi$ into Euler's Formula:

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + i(0) = -1$$

Rearranging gives Euler's Identity:

$$e^{i\pi} + 1 = 0$$

5 One of Many Applications of Euler's Equation (Signal Processing)

Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ finds numerous applications across science and engineering. Rather than listing many, we focus on one powerful application: **signal processing**. This field demonstrates how Euler's formula serves as a fundamental bridge between time-domain signals and their frequency-domain representations, enabling everything from audio processing to wireless communications.

5.1 Complex Exponentials as Building Blocks

Euler's formula is fundamental to signal processing because it allows us to represent real signals (sine and cosine waves) using complex exponentials, which are mathematically simpler to manipulate.

From Real Signals to Complex Representation

Any real sinusoidal signal can be expressed as a combination of complex exponentials:

$$\cos(\omega t) = \frac{e^{i\omega t} + e^{-i\omega t}}{2}, \quad \sin(\omega t) = \frac{e^{i\omega t} - e^{-i\omega t}}{2i}$$

where ω is the angular frequency and t is time.

This representation simplifies operations like differentiation and integration. For example:

$$\frac{d}{dt} e^{i\omega t} = i\omega e^{i\omega t}, \quad \text{while} \quad \frac{d}{dt} \cos(\omega t) = -\omega \sin(\omega t)$$

The complex exponential form keeps the same functional form after differentiation, making calculus operations cleaner.

Signal Decomposition: The Fourier Transform

The Fourier transform uses Euler's formula to decompose signals into their frequency components. For a signal $x(t)$, its Fourier transform is:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$

The inverse transform reconstructs the signal:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$$

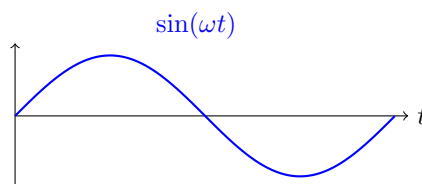
This means any signal—whether it's audio, images, or sensor data—can be represented as a sum (or integral) of complex exponentials of different frequencies.

Visualizing Signal Construction

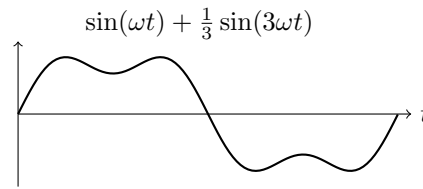
Consider building a square wave (common in digital electronics) from sine waves. Using Fourier series:

$$\text{Square wave}(t) = \frac{4}{\pi} \left[\sin(\omega t) + \frac{1}{3} \sin(3\omega t) + \frac{1}{5} \sin(5\omega t) + \dots \right]$$

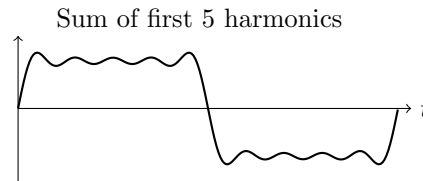
1. Fundamental sine wave (First harmonic):



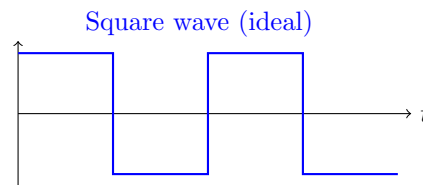
2. Adding third harmonic:



3. Sum of first 5 harmonics:



4. Square wave (result of infinite harmonics):



Each graph shows how adding more sine wave components (harmonics) approximates the square wave better. This demonstrates how complex signals are built from simple sinusoidal components using Euler's formula.

Practical Applications in Signal Processing

- **Audio processing:** Equalizers use Fourier analysis to boost/cut specific frequencies. Euler's formula enables the mathematical transformation between time-domain audio signals and their frequency spectra.
- **Image compression:** JPEG uses discrete cosine transform (a real-valued cousin of the Fourier transform) to compress images by representing them as sums of cosine waves of different frequencies.
- **Communications:** Modulation schemes like QAM (Quadrature Amplitude Modulation) use complex numbers to represent signals. The real and imaginary parts correspond to two independent data streams transmitted simultaneously.
- **Filter design:** Complex exponentials help design filters that isolate specific frequency bands. Engineers use the frequency response of filters, which is naturally expressed using Euler's formula.
- **Radar and sonar:** These systems use Fourier analysis to determine the distance and velocity of objects by analyzing reflected wave frequencies.

Euler's formula thus provides the mathematical bridge between the time domain (real signals) and frequency domain (complex representations), making it indispensable in modern signal processing. This single application demonstrates how a seemingly abstract mathematical relationship becomes a practical tool that powers our digital world—from streaming music to wireless communications and medical imaging.

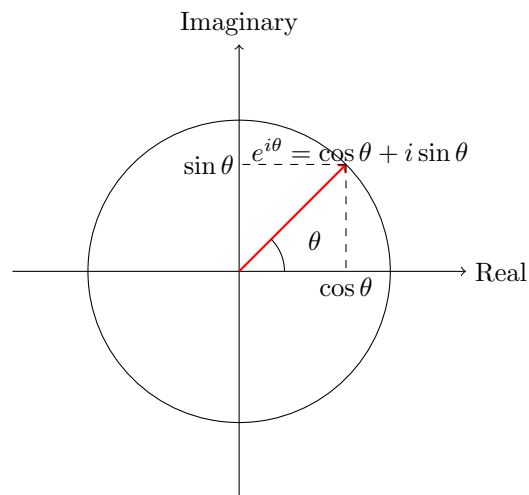
6 Visualizations and Geometric Interpretation

6.1 Complex Plane Representation

Euler's Formula can be visualized on the complex plane, where:

- Real part represents x-coordinate
- Imaginary part represents y-coordinate
- $e^{i\theta}$ represents a point on the unit circle at angle θ
- Multiplication by $e^{i\theta}$ corresponds to rotation by θ radians

6.2 Unit Circle Interpretation



7 Conclusion

Euler's Identity shows remarkable synthesis of mathematics:

- It unites 5 basic constants of various fields of mathematics.
- It shows how intimately exponential functions and trigonometric functions are connected.
- It gives an interface between real and complex analysis.
- It finds far reaching applications in physics and engineering.

Not only is it beautiful, but the fact of how $e^{i\pi} + 1 = 0$ shows the wholeness and interdependence of mathematics. It is a monument to the strength of mathematical abstractions and the beauty that can be created out of what seems to be a simple relationship.